

Python Logic Building Practice Questions

Level 1: Basic Logic

1. Even or Odd
2. Largest of Three Numbers (without max())
3. Leap Year Checker
4. Grade Calculator
5. Positive, Negative, or Zero

Level 2: Loops

1. Sum of First N Numbers
2. Factorial
3. Multiplication Table
4. Reverse a Number
5. Count Digits
6. Sum of Digits

Level 3: Number Logic

1. Armstrong Number
2. Palindrome Number
3. Prime Number
4. Print Prime Numbers (1-100)
5. Fibonacci Series
6. GCD and LCM

Level 4: Pattern Logic

1. Half Pyramid (*)
2. Reverse Half Pyramid
3. Full Pyramid
4. Number Triangle
5. Reverse Number Triangle

Level 5: Strings

1. Count Vowels
2. Reverse a String (without slicing)
3. Palindrome String
4. Character Frequency
5. Remove Duplicate Characters

Level 6: Lists

1. Find Largest Element
2. Second Largest Number
3. Remove Duplicates
4. Count Even and Odd Numbers
5. Rotate List by One Position

Level 7: Dictionaries

1. Student Marks - Find Topper
2. Word Frequency Counter

Level 8: Functions

1. Calculator using Functions
2. Recursive Factorial
3. Recursive Fibonacci

Level 9: Mini Projects

1. Guess the Number Game
2. ATM Menu System
3. Student Result Management

Bonus Challenges

1. Find Missing Number
2. Move Zeros to End
3. Find Duplicate Elements
4. Merge Two Sorted Lists
5. Check Anagrams
6. Longest Word in Sentence
7. Count Consecutive Characters
8. Rotate String by K
9. Password Validator
10. Second Most Frequent Character

Recommended Problem-Solving Approach

- Understand the problem.
- Solve sample inputs manually.
- Write the algorithm or pseudocode.
- Implement the solution in Python.

- Test with normal and edge cases.
- Optimize and refactor if possible.